Electric Circuits

ELEC 221 – Electrical Circuits

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Why Study Electric Circuits?

Electrical systems are almost found ***everywhere*** and are used by ***everybody everyday***.

- Communication Systems
- Computer Systems
- Control Systems
- Power Systems
- Signal Processing Systems

Communication Systems

Systems to generate, transmit, and distribute information:

- Telephone and data networks
- Radio telescopes
- Radar systems
- Television systems

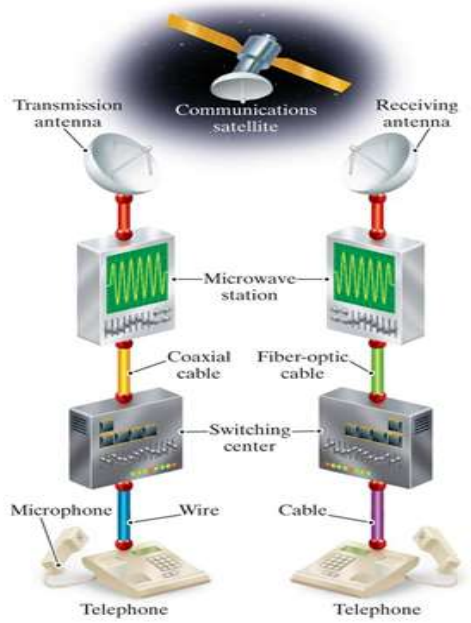


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Computer Systems

Systems to process different types of information:

- Calculators
- Smart phones and personal digital assistants
- Personal computers
- Supercomputers
- Tablets



Control Systems

Systems to regulate and control a variety of processes:

- Oil refinery
- Airplane auto pilot system
- Elevators
- Automobile fuel injection systems
- Programmable logic controllers (PLC)



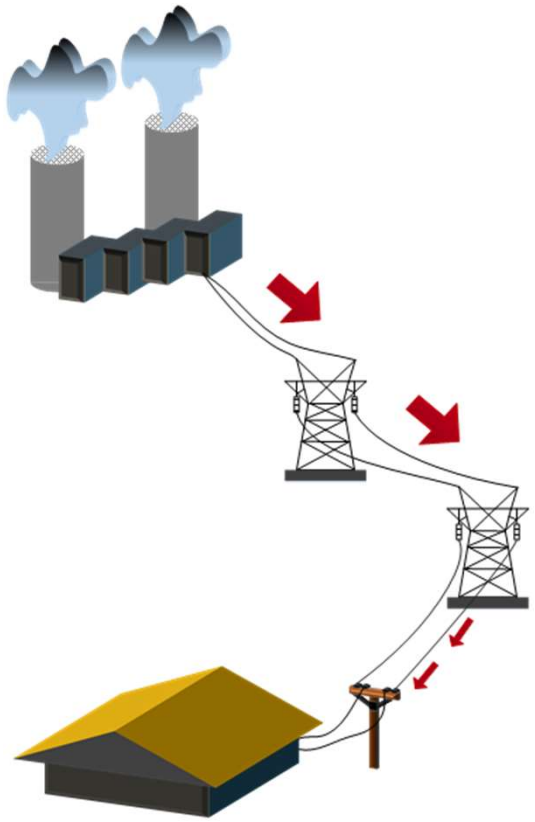
Figure: 01-03

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Power Systems

Systems to generate, transfer, and distribute electric power:

- Power generators
- Transmission lines
- Batteries
- Motors



Signal Processing Systems

Systems to reshape and transform signals and the information contained in them:

- CT scan
- MRI
- Ultrasound
- TV/VCRs
- Stereos
- Telecom systems

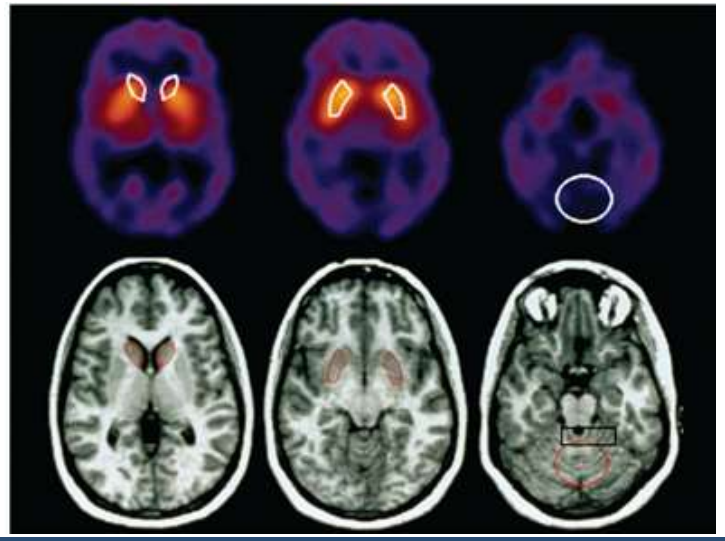


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Learning Outcomes

By the end of this video, you should be able to learn:

- **Basic Circuit Components and Theory**
  - Circuit theory
  - Circuit Analysis
  - International System of Units

Circuit Theory (cont.)

❖ **What is an "Electric circuit"?**

Mathematical model that approximates the behavior of an actual electric system.

❖ **How can electric circuits be analyzed?**

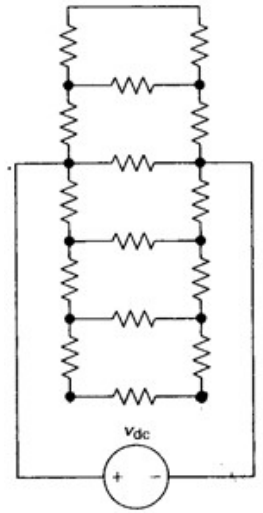
Using "Electromagnetic Field Theory" (Study of static and moving charges)

❖ **Why Circuit Theory?**

Electromagnetic field theory is too involved.

❖ **What is Circuit Theory?**

The analysis tools and laws that will be discussed in this course (kirchhoff's laws, Ohm's law, Thevenin and Norton theorems,…)



Fi g 1-3. Circuit model of the defroster grid.

Circuit Theory (cont.)

- We are lucky that under certain assumptions, one can analyze certain electrical systems and circuits using the much simpler circuit theory

- Assumptions necessary for using Circuit Theory?

  1. The circuit can be considered as "lumped parameter", i.e. the electrical effects happen instantaneously throughout the physical circuit. (be small enough!)
  2. The net charges on every component in the system is always zero.
  3. There is no magnetic coupling between the components is a system.

What does “small enough” mean?

The rule of 1/10th:

If circuit is smaller than 1/10th of its wavelength, it can be considered a lumped-parameter system.



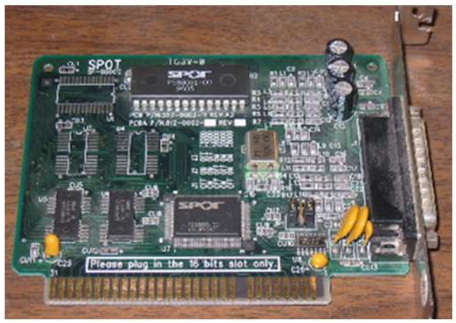
Reminder:

Electric signals travel at near the speed of light. Thus, for a 60 Hz power grid:

$$\lambda = \frac{c}{f} = \frac{3\times10^{8}\ \text{m/s}}{60\ \text{Hz}} = 5\times10^{6}\ \text{m} \quad \Rightarrow \quad \text{Circuit} < 500\ \text{km}$$

Alternately, at radio frequencies we have:

$$\lambda = \frac{c}{f} = \frac{3\times10^{8}\ \text{m/s}}{10^{9}\ \text{Hz}} = 0.3\ \text{m} \quad \Rightarrow \quad \text{Circuit} < 3\ \text{cm}$$



Circuit Analysis

What is “circuit analysis”?

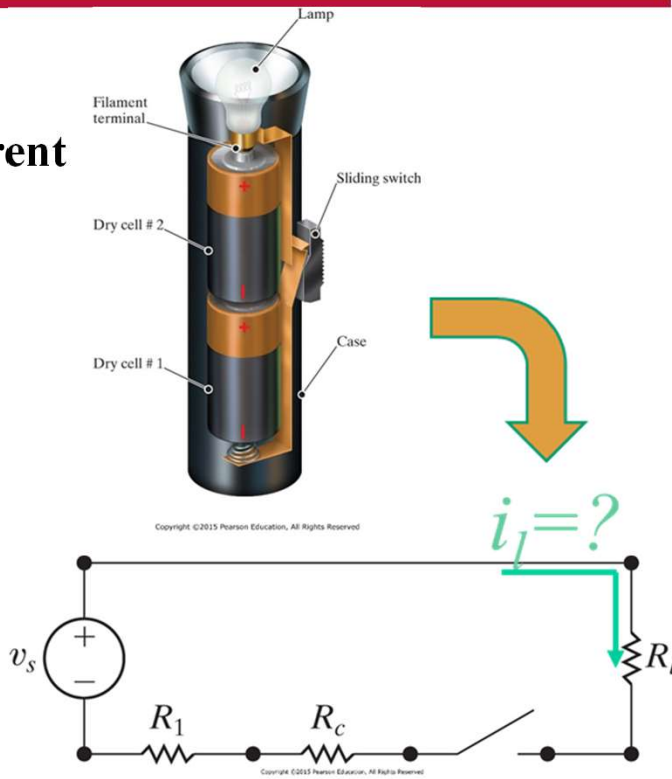
- Understanding the behavior of **each component** in terms of its **current** (i) and **voltage** (v) in a circuit model.

What is a “circuit model”?

- A mathematical model of a physical electric system.
- Made of “ideal basic circuit elements”.

Note the **differences** between:

- a circuit model and a physical electric system
- circuit analysis and measurements of a physical electric system



International System of Units

TABLE 1.1 The International System of Units (SI)

Quantity	Basic Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	degree kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

National Institute of Standards and Technology Special Publication 330, 2008 Edition, Natl. Inst. Stand. Technol. Spec. Pub. 330, 2008 Ed., 96 pages (March 2008)

International System of Units

TABLE 1.2 Derived Units in SI

Quantity	Unit Name (Symbol)	Formula
Frequency	hertz (Hz)	s^{-1}
Force	newton (N)	$kg \cdot m/s^2$
Energy or work	joule (J)	$N \cdot m$
Power	watt (W)	J/s
Electric charge	coulomb (C)	$A \cdot s$
Electric potential	volt (V)	J/C
Electric resistance	ohm (Ω)	V/A
Electric conductance	siemens (S)	A/V
Electric capacitance	farad (F)	C/V
Magnetic flux	weber (Wb)	$V \cdot s$
Inductance	henry (H)	Wb/A

National Institute of Standards and Technology Special Publication 330, 2008 Edition, Natl. Inst. Stand. Technol. Spec. Pub. 330, 2008 Ed., 96 pages (March 2008)

TABLE 1.3 Standardized Prefixes to Signify Powers of 10

Prefix	Symbol	Power
atto	a	10^{-18}
femto	f	10^{-15}
pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
deka	da	10
hecto	h	10^{2}
kilo	k	10^{3}
mega	M	10^{6}
giga	G	10^{9}
tera	T	10^{12}

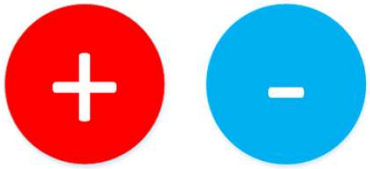
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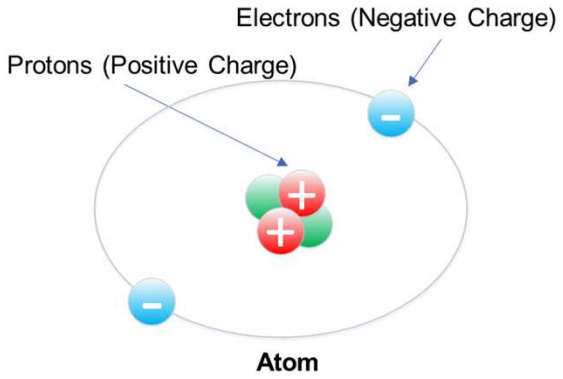
- **Basic Circuit Components and Theory**
  - Explain voltage and current relations.
  - Ideal basic circuit component

Electric Charge

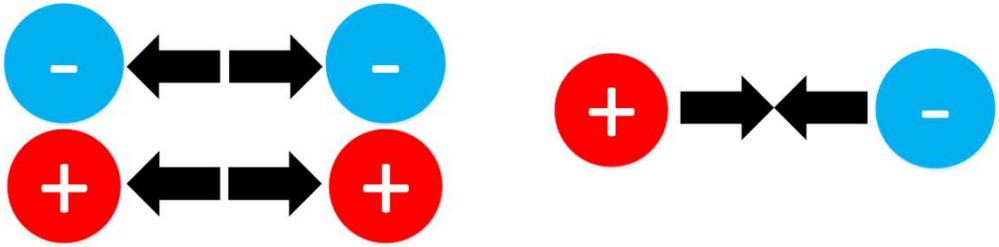
Electric Charge

- Two types → ***Positive*** and ***Negative***





- ***Like charges repel, opposite charges attract***

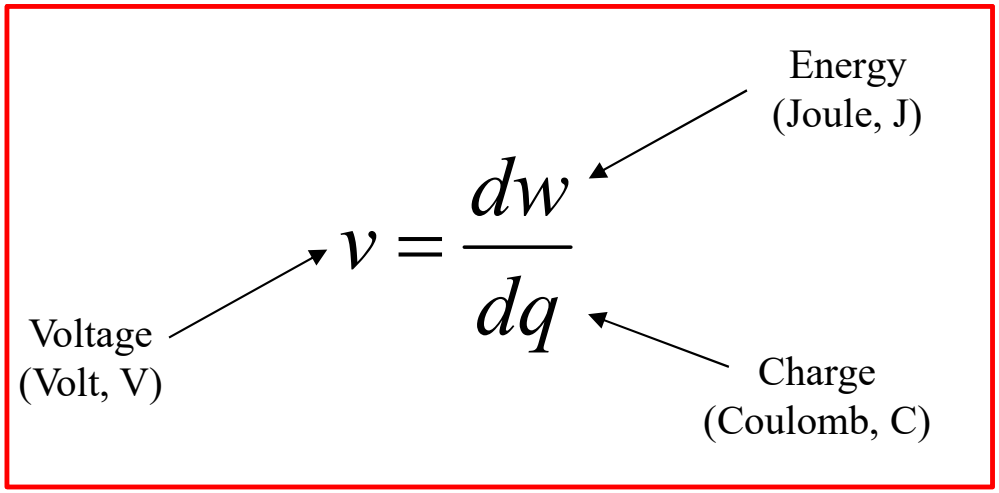


Electric Charge, continued

- Electric charge is the basis for describing all electric phenomena
- Properties of electric charge:
  - ✓ It is bipolar (positive and negative charges)
  - ✓ Exists in discrete quantities
    - ➢ i.e. integral multiples of an electron charge: e = 1.6022 x 10-19 C
  - ✓ Electrical effects result from:
    - ➢ Separation of Charge: voltage
    - ➢ Flow of Charges: current

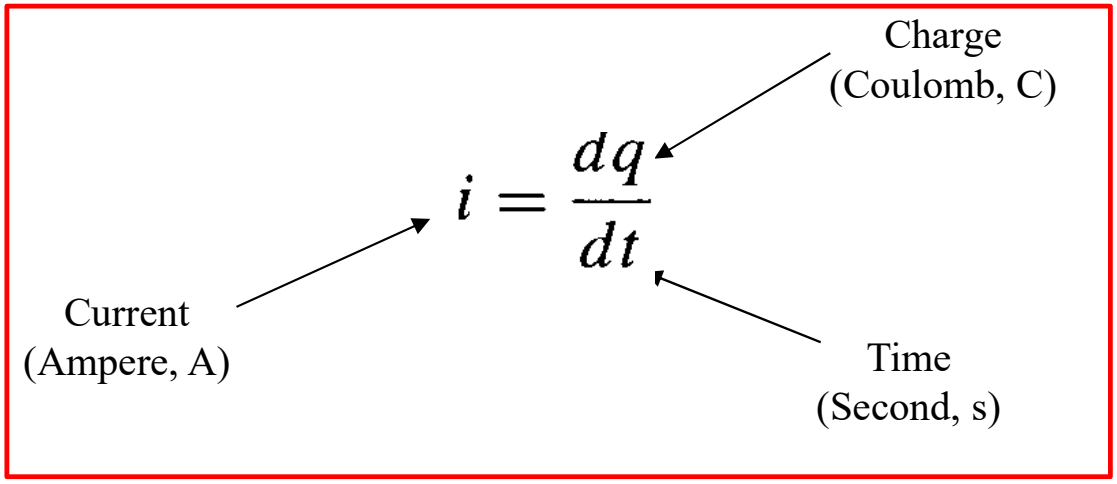
Voltage

Definition: Voltage is the energy required per unit of charge for separation.



Current

Definition: Current is the rate of charge flow



- Current is considered as continuous quantity although it is made up of discrete components.
- The conventional current flow represents the movement of positive charges.

Voltage and Current



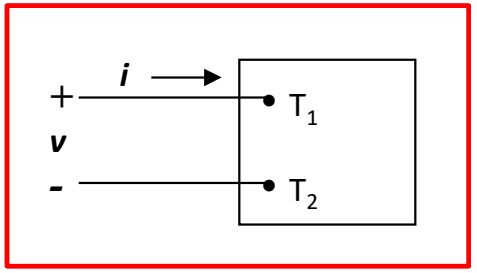
- In circuit theory, we will be interested in how the current through an element is changing in relation to its terminal voltage and not in the internal behavior.

- We will use circuit models already developed and will not try to model elements.

- Modeling requires knowledge of the physics of the device: advanced solid state and electronics (later).

Ideal Basic Circuit Element

Properties:

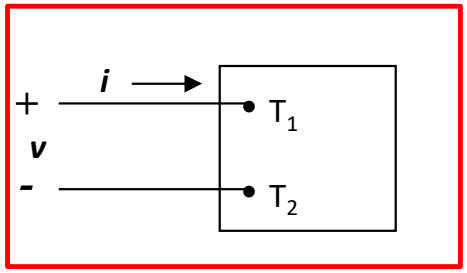


- It has two terminals.
- It is described mathematically in terms of its current (i) and voltage (v)
  - Voltage polarity reference: + -
  - Current direction reference: ⟶
  - References are chosen arbitrarily.
- It cannot be subdivided into other elements.
- Why ideal basic?
  - "**Ideal**" since it is does not exist as a physically realizable component. Ideal elements can be connected to model an actual device.
  - "**Basic**" since it cannot be further reduced or subdivided.

Ideal Basic Circuit Element (cont.)

❖ **Voltage across an element:**

The energy required to move a positive charge of 1 C through an element



Example:

Determine the energy required to move 120 C of charge from T_2 to T_1 if v = +2 V

$$v = \frac{dw}{dq} \Rightarrow w = \int_0^q v\,dx = vq = 2\,\text{V} \times 120\,\text{C} = 240\,\text{J}$$

Ideal Basic Circuit Element (cont.)

Current through an element:

The rate of flow of charge:

Example: If the expression for the charge entering the upper terminal is:

$$q = \frac{1}{\alpha^2} - \left(\frac{t}{\alpha} + \frac{1}{\alpha^2}\right)e^{-\alpha t}\ \text{C}$$

Find the maximum value of the current entering the terminal if $\alpha = 0.03679$ s-1.

$$i = \frac{dq}{dt} = \frac{d}{dt}\frac{1}{\alpha^2} - \frac{d}{dt}\left(\frac{t}{\alpha} + \frac{1}{\alpha^2}\right)e^{-\alpha t}\ \text{C}$$

$$= 0 - \frac{d}{dt}\left(\frac{t}{\alpha} + \frac{1}{\alpha^2}\right)e^{-\alpha t}\ \text{C}$$

$$= -\frac{1}{\alpha}e^{-\alpha t} + te^{-\alpha t} + \frac{1}{\alpha}e^{-\alpha t}$$

$$= te^{-\alpha t}$$

$$\frac{di}{dt} = \frac{d}{dt}te^{-\alpha t}$$

$$= e^{-\alpha t} - \alpha te^{-\alpha t} = 0$$

$$e^{-\alpha t} = \alpha te^{-\alpha t}$$

$$t_{\max} = 1/\alpha$$

$$i_{\max} = \alpha^{-1}e^{-1} = 10\ \text{A}$$

Voltage and Current

V > 0 ⟺ Voltage drops from T_1 to T_2

⟺ Voltage rises from T_2 to T_1

(terminal with + sign has higher potential)

V < 0 ⟺ Voltage rises from T_1 to T_2

⟺ Voltage drops from T_2 to T_1

(terminal with + sign has lower potential)

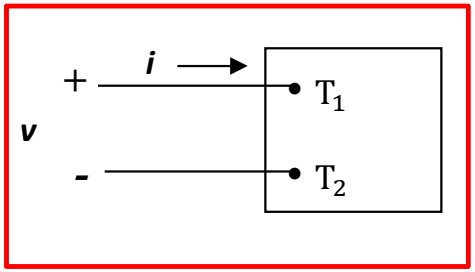
i > 0 ⟺ Positive charge flows from T_1 to T_2

⟺ Negative charge from T_2 to T_1

i < 0 ⟺ Negative charge flows from T_1 to T_2

⟺ Positive charge from T_2 to T_1

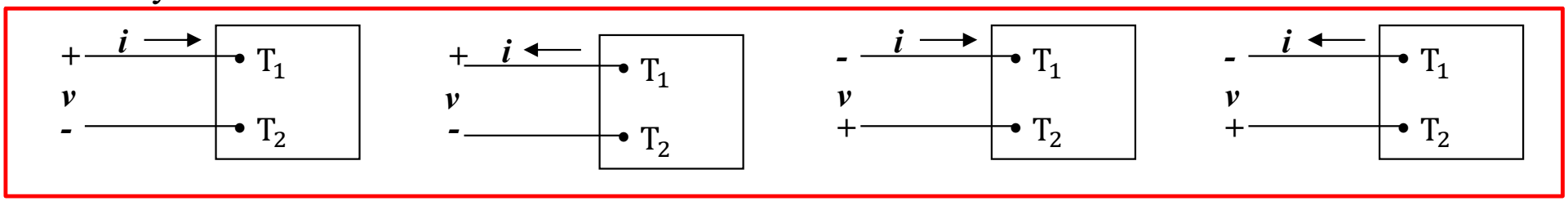
Only for below reference assignment



- **Basic Circuit Components and Theory**
  - Power: Passive sign convention.
  - Energy conservation.

Interpretation of Reference Polarity and Directions

➢ The assignment of the reference polarity for voltage and the reference direction for current is arbitrary:



➢ The sign of the value of v determines whether T_1 has higher potential energy than T_2 or vise-versa.

➢ The sign of the value of i determines in which direction positive charges are moving.

v = +2 V	v = +2 V	v = -2 V	v = -2 V
i = +1 A	i = -1 A	i = +1 A	i = -1 A

➢ No matter what voltage polarity and current direction reference we choose, all of the above represent the same physical fact:

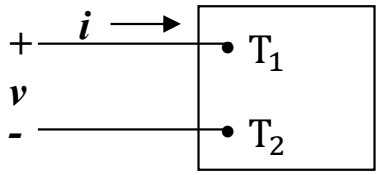
$VT_1 = VT_2 + 2$ & positive charge enters T_1

Passive Sign Convention (PSC)

Question: What sign convention should we pick for the subsequent equations such that the equations describe the same physical phenomenon regardless of the chosen voltage polarity and current direction references?

Passive Sign Convention: Whenever the **current reference direction** enters the terminal with positive (negative) **voltage reference polarity** use positive (negative) sign in any expression that relates the voltage to current.

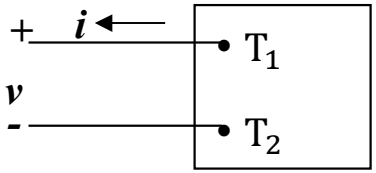
Example: Resistor (R = 2 Ω)



i = +1

v = + R i

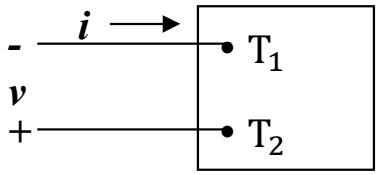
(+2) = + (2) (+1)



i = -1

v = - R i

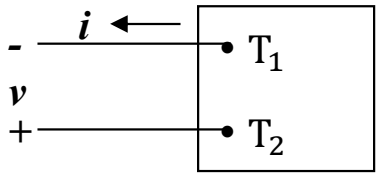
(+2) = - (2) (-1)



i = +1

v = - R i

(-2) = - (2) (+1)



i = -1 A

v = + R i

(-2) = + (2) (-1)

Power and Energy

Why power & energy calculation is important?

- In practice the useful outputs of the many systems are non-electric and power and energy can make the link between electric and non-electric measures and performance indices.

- Practical devices have limitation on the amount of power that they can deliver.

Power and Energy (cont.)

***Definition*:** Power is the time rate of expending or absorbing energy

$$p = \frac{dw}{dt}$$

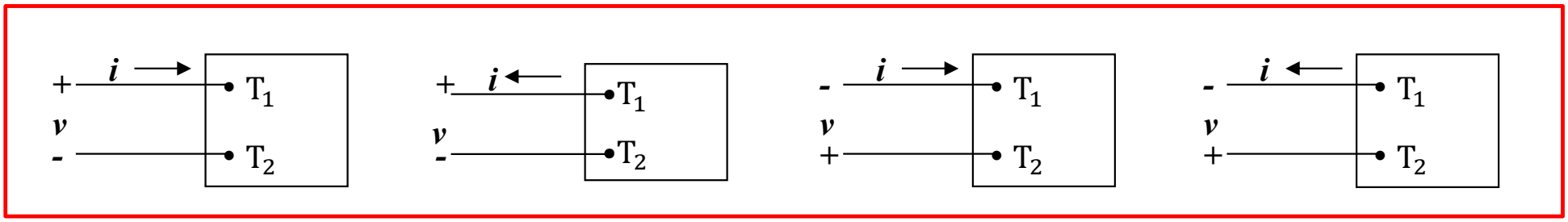
Power (Watt, W) — Energy (Joule, J) — Time (Second, s)

P > 0 power is delivered to the element
power is absorbed by the element

P < 0 power is extracted from the element
power is generated in the element

$$p = \frac{dw}{dt} = \left(\frac{dw}{dq}\right)\left(\frac{dq}{dt}\right) = vi$$

Power & Energy (Use of passive sign convention)



$i = +1\ A$	$i = -1\ A$	$i = +1\ A$	$i = -1\ A$
$v = +Ri$	$v = -Ri$	$v = -Ri$	$v = +Ri$
$(+2) = +(2)\ (1)$	$(+2) = -(2)\ (-1)$	$(-2) = -(2)\ (+1)$	$(-2) = +(2)\ (-1)$
$\boldsymbol{P = +vi}$	$\boldsymbol{P = -vi}$	$\boldsymbol{P = -vi}$	$\boldsymbol{P = +vi}$
$(+2) = +(+2)(1)$	$(+2) = -(+2)(-1)$	$(+2) = -(-2)(+1)$	$(+2) = +(-2)(-1)$

Example: Power and Energy

Example:

When a car has a dead battery, it can often be started by connecting the battery from another car across its terminals. The positive terminals are connected together as are the negative terminals. The connection is illustrated in the figure. Assume current i in the figure is measured and found to be 30 A:

1. Which car has the dead battery?
2. If this connection is maintained for 1 minute, how much energy is transferred to the dead battery?

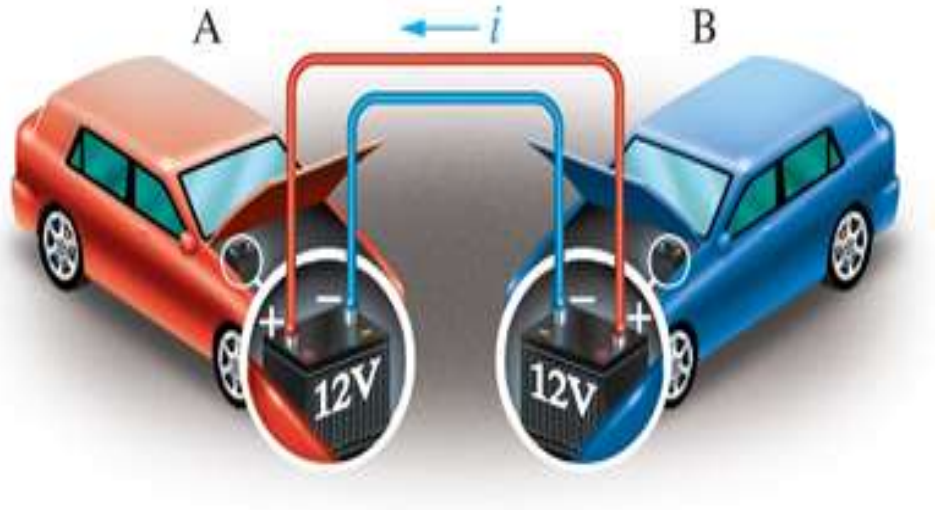
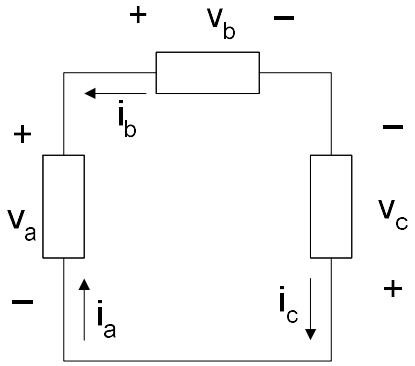


Fig. 1-4. Copyright © 2015 Pearson Education, All Rights Reserved.

Principle of Conservation of Energy

❖ Total power delivered = total power absorbed

$$\sum p = 0$$



Problem: Find v_b and v_c if

$v_a = 10$ V, $p_b = 9$ W,
$i_a = 1$ A, $i_b = -1$ A, $i_c = 1$ A

$$p_a = -vi \quad \text{(PSC)}$$
$$= -(10)(1) = -10$$

$$0 = p_a + p_b + p_c \quad \text{(CoE)}$$
$$\Rightarrow p_c = -p_a - p_b = 1 \text{ W}$$

$$v_b = -p_b / i_b$$
$$= -9/(-1) = 9 \text{ V}$$

$$v_c = -p_c / i_c$$
$$= -1/1 = -1 \text{ V}$$

Conservation of Power (cont.)

Conservation of power in fact is also a direct product of KVL and KCL.

